Complete EFX for Four Additive Agents and Eight Goods

A formally audited computer-assisted proof

Draft for adversarial review

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Abstract

We prove that every instance of four agents and eight indivisible goods with nonnegative additive valuations admits a complete allocation that is envy-free up to any good (EFX). This is the first additive parameter cell immediately beyond the general $m \leq n + 3$ existence frontier. The proof is hybrid. Lean verifies the mathematical reduction from a hypothetical counterexample to eight exact rational linear-arithmetic systems. A generator-independent parser audit binds every assertion in those systems to the Lean terminal-cell specifications. For each system, cvc5 emits a CPC proof of unsatisfiability and the separate Ethos checker accepts the proof as `correct`. The eight proofs contain 13,812,082 proof-step lines and occupy 1.668 GB. An alternative encoding over all 40,824 labeled onto allocations independently closes all eight cells. A 124-entry SHA-256 manifest binds the complete proof package, exact published inputs, tools, audits, formulas, proofs, Lean build, and source archive. We explain the structural reduction, the formal-to-computational boundary, and the remaining trust assumptions. We also report a 130-partition deletion-minimal explanatory core with its own independently checked CPC proof. This manuscript is a technical review draft, not a claim of peer review or independently confirmed novelty.

1 Introduction

Let $\mathcal{N}$ be a finite set of agents and $\mathcal{M}$ a finite set of indivisible goods. Agent i has an additive valuation

$$v_i(S) = \sum_{g \in S} w_{ig}, \qquad w_{ig} \geq 0.$$

A complete allocation $X = (X_i)_{i \in \mathcal{N}}$ partitions all goods. It is EFX if for every pair of agents $i \neq j$ and every good $g \in X_j$,

$$v_i(X_i) \geq v_i(X_j \setminus \{g\}).$$

The central existence question asks whether a complete EFX allocation always exists for additive valuations.

Complete EFX is known for three additive agents [1]. Mahara proved complete EFX for general monotone valuations when $m \leq n + 3$ [2]. Thus four agents and seven goods are covered, while the case of four agents and eight goods is the first near-diagonal additive cell outside both standard boundaries. Four-agent partial EFX leaving at most one good unallocated is known [3], but does not settle completeness.

Our main result is the following.

Theorem 1.1 (Main theorem)**.** *Every nonnegative additive valuation instance with four agents and eight goods admits a complete EFX allocation.*

The terminal reduction is carried out for strictly positive weights. To pass from nonnegative weights, observe that each fixed allocation's EFX region is closed and there are finitely many allocations. Hence the set of instances with no EFX allocation is open. Any nonnegative counterexample would therefore have a sufficiently small strictly positive perturbation that remains a counterexample. This perturbation reduction and the final nonnegative wrapper are checked in Lean.

The proof combines four layers:

1. published complete-EFX inputs for four agents and seven goods and for three additive agents;
2. a Lean-checked structural reduction to eight terminal ranking cells;
3. an independently generated and parser-audited exact linear-arithmetic formula for each cell; and
4. eight cvc5 CPC proofs accepted by the separate Ethos checker.

This organization follows an explicit trust boundary. The mathematical reduction is not inferred from solver output. Conversely, Lean does not prove billions of arithmetic proof steps internally. It proves the finite reduction; exact formula audits connect that reduction to independently checked arithmetic certificates.

Contributions. Besides Theorem 1.1, the development provides:

- a reverse-parking criterion and a terminating threshold-cycle normal form for EFX allocations;
- a generic favorite-small-bundle theorem: if every agent receives a favorite good and every bundle has size at most two, the allocation is EFX;
- a formal construction closing the injective-favorite case whenever $n \le m \le 2n$;
- exact canonical coverage, stabilizer, partition-enumeration, and perfect-matching equivalence theorems in Lean;
- an independent direct-allocation encoding for all eight cells; and
- a deletion-minimal 130-partition core that exposes the geometry of one terminal contradiction.

2 Definitions and basic reductions

Definition 2.1. For an agent i and bundle S, define the one-good-removal threshold

$$\tau_i(S) = \max_{g \in S} v_i(S \setminus \{g\}), \qquad \tau_i(\varnothing) = 0.$$

An allocation is EFX exactly when

$$v_i(X_i) \ge \tau_i(X_j) \quad \text{for all } i, j.$$

For additive valuations,

$$\tau_i(S) = v_i(S) - \min_{g \in S} v_i(g).$$

2.1 Reverse parking and threshold cycles

Suppose X is EFX and $f \in X_h$. Removing f can only make the observer inequalities of h harder; all other agents see a target bundle shrink. Consequently f is removable from h precisely when

$$v_h(X_h) - v_h(f) \geq \max_{j \neq h} \tau_h(X_j).$$

Choosing a least-valued good in X_h yields the following criterion.

Lemma 2.2 (Reverse parking)**.** *An owner h has a removable good if and only if*

$$\tau_h(X_h) \geq \max_{j \neq h} \tau_h(X_j).$$

If no owner satisfies this inequality, choose for every owner h a strict threshold-maximizing target $c(h)$. The resulting functional graph contains a directed cycle. Rotate bundles along that cycle. EFX is preserved, and the finite potential

$$\Phi(X) = \sum_h \tau_h(X_h)$$

strictly increases. Repeating must terminate.

Theorem 2.3 (Threshold-cycle normal form)**.** *Every complete EFX allocation with nonempty bundles can be transformed by EFX-preserving bundle rotations into one having an owner-removable good.*

The theorem is generic over finite agents and goods and nonnegative additive valuations in the Lean development. Strict positivity forces all four bundles to be nonempty in the 4×8 setting.

2.2 Normalization and strict rankings

Multiplying one agent's valuation row by a positive scalar preserves every EFX comparison. We normalize each row so that

$$\sum_{g \in \mathcal{M}} w_{ig} = 1.$$

There are only finitely many complete allocations. If a normalized instance has no EFX allocation, choose one strict failed inequality for each allocation. These failures persist in an open neighborhood. A sufficiently small sum-preserving perturbation therefore makes every row ranking strict while preserving positivity and nonexistence.

Lean proves this simultaneous perturbation rather than treating genericity as an informal "without loss of generality" step.

2.3 Favorite small bundles

Lemma 2.4 (Favorite-small-bundle lemma)**.** *Let X be a complete allocation in a positive additive instance. If every agent owns one of its favorite goods and every bundle has size at most two, then X is EFX.*

Proof. After removing any good from a target bundle, at most one good remains. The observer values that good no more than its own favorite, which is contained in its own bundle. □

The Lean theorem is more general: given an injective favorite map and $|\mathcal{M}| \leq 2|\mathcal{N}|$, it constructs the allocation and proves EFX. In particular, if four agents have four distinct favorite goods among eight goods, give each agent its favorite and one remaining good.

3 Reduction to eight terminal cells

Assume for contradiction that a positive additive 4×8 instance has no complete EFX allocation. Normalize and perturb it as above.

3.1 Residual cuts from published theorems

Two structurally easy regions are removed.

Universal minimum. If a good is minimum for every agent, delete it, apply Mahara's complete 4×7 theorem, rotate to an unenvied source, and insert the good there.

Peelable agent-good pair. If an agent-good pair satisfies the formal peelability inequalities, delete that agent and good, apply the complete three-agent additive theorem to the remaining seven goods, and reinsert the pair.

The latter reduction initially exposed a genuine specification error during audit: the deleted agent had incorrectly been retained as an observer with an empty bundle. The corrected predicate excludes that observer while retaining all comparisons needed for reinsertion. The correction and all downstream theorems are machine-checked.

3.2 Favorite multiplicities

By Lemma 2.4, favorites cannot be distinct. With strict rankings, the noninjective favorite multiplicities are, up to relabeling,

$$2+1+1, \qquad 3+1, \qquad 2+2, \qquad 4.$$

Canonical favorite vectors for the first three are

$$(0,0,1,2), \qquad (0,0,0,1), \qquad (0,0,1,1).$$

If all four agents favor good 0, refine by the multiplicities of their second favorites:

$$1+1+1+1, \qquad 4, \qquad 3+1, \qquad 2+2, \qquad 2+1+1.$$

This gives exactly eight terminal cells.

Only symmetries preserving a fixed ranking prefix are used. Agents with the same prefix and goods absent from every prescribed prefix may be sorted lexicographically. Lean proves that every relevant orbit has a representative satisfying these stabilizer inequalities.

4 Partitions and perfect matchings

There are exactly

$$S(8,4) = 1,701$$

unlabeled partitions of eight goods into four nonempty bundles. The Lean development verifies the restricted-growth enumeration and proves that every onto labeled allocation is obtained from one enumerated partition by a permutation of the four bundles.

Fix a partition $\mathcal{B} = (B_0, B_1, B_2, B_3)$. Construct a bipartite graph between agents and bundles. Put edge iB_b when

$$v_i(B_b) \geq \max_{c \neq b} \tau_i(B_c).$$

Proposition 4.1 (Matching equivalence)**.** *The partition $\mathcal{B}$ supports a complete EFX allocation if and only if its agent-bundle feasibility graph has a perfect matching.*

Thus a counterexample must obstruct all $4! = 24$ matchings for every one of the 1,701 partitions.

5 Exact terminal formulas

Each terminal formula is quantifier-free rational linear arithmetic. It contains:

- 32 strictly positive valuation variables and four row-normalization equations;
- exact favorite or second-favorite constraints;
- the valid cell stabilizers;
- exclusions of the universal-minimum and peelable regions;
- 27,216 agent-bundle feasibility-edge definitions; and
- 40,824 perfect-matching obstruction assertions.

The final formulas are generated by a program that does not import the original encoder, the original partition implementation, or Z3. A separate parser audit reconstructs the exact assertion semantics, independently enumerates the 1,701 partitions, checks every edge definition and all 24 matching obstructions per partition, and binds the cell constants to their Lean definitions.

Cell	Ranking data	Stabilizer
favorite-211	$(0, 0, 1, 2)$	none
favorite-31	$(0, 0, 0, 1)$	favorite preserving
favorite-22	$(0, 0, 1, 1)$	favorite preserving
second-1111	$(1, 2, 3, 4)$	prefix preserving
second-4	$(1, 1, 1, 1)$	prefix preserving
second-31	$(1, 1, 1, 2)$	prefix preserving
second-22	$(1, 1, 2, 2)$	prefix preserving
second-211	$(1, 1, 2, 3)$	prefix preserving

Table 1: The eight terminal ranking cells. The last five share favorite vector $(0, 0, 0, 0)$.

6 Certificate closure

For every terminal formula:

1. the original exact Z3 computation returns `unsat`;
2. the archived formula replays in a fresh process;
3. an isolated rebuild matches the archived formula byte for byte;
4. the independent generator emits a semantically audited formula;
5. cvc5 emits a CPC proof; and
6. Ethos accepts that proof as `correct`.

The eight CPC proofs total 1,668,457,899 bytes and 13,812,082 proof-step lines. Formula, proof, tool, all 51 transitive CPC-signature files, paper, and audit hashes are bound by a 96-entry certificate manifest. An additional audit matches all 545,360 formula assertions to all 545,360 permanent CPC assumptions in order, with zero mismatches.

6.1 Alternative direct encoding

As a non-load-bearing cross-check, a second formulation directly negates every labeled onto allocation. There are

$$4!\,S(8,4) = 40,824$$

such allocations. This encoding does not use the 1,701-partition feasibility-graph representation. All eight direct formulas are UNSAT: six close in Z3, and the two hardest close in cvc5 after an audited identity-only SMT-LIB syntax normalization.

6.2 Lean composition and axiom audit

Lean proves the theorem

`admitsEFX_4x8_of_standard_published_inputs`.

Its permanent axiom audit reports exactly

[`propext`, `Classical.choice`, `Quot.sound`],

with no `sorryAx` dependency and no native-computation axiom. The exact Mahara and Chaudhury–Garg–Mehlhorn paper versions and theorem fragments are hash-pinned; their proofs remain external trusted mathematics rather than re-formalized Lean proofs.

6.3 End-to-end package

The frozen finalizer:

- verifies every CPC and direct-encoding artifact hash;
- reruns the semantic and published-input audits;
- checks the Lean source inventory and root imports;
- performs a full pinned Lean build and axiom audit; and
- creates a deterministic archive containing the transitive proof sources.

All 124 entries in the final SHA-256 manifest verify. The deterministic source archive contains 72 files. A clean-source replay script extracts it into a new temporary project with no prior project build directory and repeats both audits, the Lean build, and the axiom check; independent-host replay remains part of the external release gate.

Bundle-size shape	Initial core	Final core
$3+2+2+1$	98	59
$2+2+2+2$	51	20
$4+2+1+1$	30	30
$3+3+1+1$	20	20
$5+1+1+1$	1	1
Total	200	130

Table 2: Shape counts in the explanatory core.

7 The 130-partition explanatory core

The favorite-211 cell has favorite vector $(0, 0, 1, 2)$. Starting from all 1,701 partition obstructions, assumption-core extraction produced 200 partitions. Deletion shrinking then obtained a 130-partition core.

Every single-partition deletion from the final core is satisfiable, so every survivor is individually indispensable relative to this core. This is deletion minimality, not a minimum-cardinality theorem.

The final core has an exact two-piece description:

1. 110 partitions in which shared favorite good 0 is a singleton, favorite goods $0, 1, 2$ are separated, and no nonfavorite good is a singleton; and

2. 20 pair partitions of shape $2+2+2+2$ in which good 0 is paired with one nonfavorite good.

The reduced formula has a 220,777,714-byte CPC proof containing 2,032,963 proof-step lines; Ethos accepts it as `correct`.

This geometry suggests an analytic target: prove that either one of the anchored singleton partitions or one of a small boundary family of anchored pair partitions has a perfect EFX matching. Ongoing deletion-witness analysis extracts exact valuations for which each indispensable partition is the sole EFX-supporting partition within the 130-partition core, counts all support in the full 1,701-partition universe, and exposes the Hall-deficient agent subsets in the other 129 core partitions.

8 Trust boundary and review status

The internally verified deduction has the form

$$\begin{aligned} &\text{two pinned published EFX theorems} \\ &+ \text{ Lean-checked structural reduction} \\ &+ \text{ independently audited Lean-to-SMT binding} \\ &+ \text{ eight CPC/Ethos UNSAT certificates} \\ &\Longrightarrow \text{ complete positive additive EFX at } (4, 8). \end{aligned}$$

The remaining trusted components are:

- the mathematical proofs in the two published input papers;
- the Lean kernel, pinned toolchain, and mathlib;
- the Ethos checker implementation; and

- the parser-based semantic bridge, which is audited software rather than a single reflected theorem in Lean's kernel.

No independent fair-division or formal-certification expert has yet signed off. The package has a fresh-source replay on the original host, but not a second-machine or cross-operating-system replay. Accordingly, the warranted status is:

> **Internally verified hybrid computer-assisted theorem; external review and novelty confirmation pending.**

9 Research directions

Three directions are immediate.

Short analytic contradiction. Use the deletion witnesses to identify repeated Hall-deficient agent sets and translate their missing edges into threshold inequalities. Search for small Farkas combinations that explain the 130-way disjunction.

Next parameter cells. The universal-minimum and peelable reductions make $(4, 9)$ and $(5, 9)$ the natural successors. At $(5, 9)$, the generic injective-favorite theorem still closes the distinct-favorite region because $9 \leq 2 \cdot 5$. Hall-selector encodings may avoid enumerating all $5! = 120$ matchings per partition.

Counting. Theorem 1.1 proves at least one EFX allocation at $(4, 8)$. The additive counting conjecture asks for at least four. Threshold-cycle rotations and multiplicities of perfect matchings may help strengthen existence into a counting statement.

Artifact availability

The source repository contains the Lean development, generators, audit programs, proof guide, expert-review checklist, and this manuscript. The large frozen formulas and proof objects are stored in the certification bundle identified by `EFX48_END_TO_END_CERTIFICATE.json` and `EFX48_END_TO_END_MANIFEST.sha256`.

References

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